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New Early and Late Carboniferous paleomagnetic results from the Qaidam Block, NW China: Implications for the paleogeography of Central Asia

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ABSTRACT

There is an ongoing debate regarding the paleogeographic evolution of the Qaidam Block (37.3 °N/96.4 °E) in northwest China during the late Paleozoic. To provide a reliable constraint on the paleogeographic reconstruction and to determine the relationship with the adjacent area, we performed a paleomagnetic investigation of Early and Late Carboniferous sediments from the Qaidam Block. Stepwise thermal demagnetization successfully isolated high unblocking-temperature characteristic directions from the samples. The tilt-corrected mean direction of the Early Carboniferous sediments is $Ds/Is = 298.6°/-46.4°$ with $\alpha_{95} = 4.7°$ and N = 15 sites, corresponding to a paleopole at −3.2 °N, 147.5 °E with $A_{95} = 4.8°$. The Late Carboniferous paleomagnetic direction is $Ds/Is = 320.1°/-46.1°$ with $\alpha_{95} = 5.3°$ and N = 8 sites, corresponding to a paleopole at −15.1 °N, 132.2 °E with $A_{95} = 5.4°$. A positive fold test for the Early Carboniferous paleomagnetic directions, and consistency with the Kiaman reversed superchron for the Late Carboniferous paleomagnetic directions, indicates the primary origin of the characteristic remanence of the studied section. The new paleomagnetic results suggest that the Qaidam Block was located at about 27 °N and did not undergo significant N-S movement during the Carboniferous. However, the Qaidam Block experienced a counter-clockwise rotation of about 21° during this period. Comparisons of the paleomagnetic results from the major blocks of Central Asia suggest that the Qaidam Block may have experienced a relatively rapid northward movement and collided with the Tarim Block after the Late Permian. Based on the paleomagnetic poles of the Qaidam Block and adjacent blocks, we present a tentative paleogeographic reconstruction for central Asia during the Early and Late Carboniferous.

1. Introduction

Central Asia comprises and was formed by the collision and amalgamation of several continents/blocks (Siberia, North China, Tarim and several microcontinents). The closure of ancient oceans and subsequent collisions of the major blocks of Central Asia formed the Central Asian Orogenic Belt (CAOB, Şengör et al., 1993; Jahn et al., 2004; Xiao et al., 2004, 2009, 2014, 2015; Windley et al., 2007; Zheng et al., 2013; Dong et al., 2016). The Qaidam Block is the major block in the southern part of Central Asia. It has been proposed that the paleo-position of the Qaidam Block was in close proximity to the Tarim Block since the Middle Devonian (Heubeck, 2001) and that it may have originally formed a part of the Tarim Block (Metcalfe, 2006, 2013). In terms of tectonic affinity, several workers have considered that the Qaidam and Tarim Blocks belong to a unique craton (Ge and Liu, 2000; Duan and Ge, 2005; Ge et al., 2009). However, Late Permian paleomagnetic results indicated that the Qaidam Block was amalgamated with the Tarim Block after the Carboniferous (Xu et al., 2011). Moreover, Cocks and Torsvik (2007, 2013) considered that the Qaidam Block formed the southwestern part of the North China Block. The principal features of Central Asia were formed by the end of the Paleozoic (Metelkin et al., 2010). Hence, the Carboniferous is an important interval for deciphering the paleogeography, tectonic transition and evolution of the Qaidam Block.

Paleomagnetism is the one of the most effective methods for reconstructing the paleo-latitudinal displacement of tectonic blocks.

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Previous Carboniferous paleomagnetic studies for the Siberia continent (Kravchinsky et al., 2002), the Tarim Block (Bai et al., 1987; Fang et al., 1996; Gilder et al., 1996), the Mongolia Block (Zhao et al., 1990; Zhao et al., 2013), the Hexi Corridor-Alashan Block (Wu et al., 1993; Huang et al., 2001; Yuan and Yang, 2015), the Junggar Block (Yi et al., 2015) and the Yili Block (Wang et al., 2007) have provided important constraints on the paleogeography and tectonic evolution of Central Asia. The Qaidam Block is a key area connecting the Tarim, North China Blocks and the Tibetan Plateau in Central Asia. However, previous paleomagnetic studies of the Qaidam Block have focused mainly on the Cenozoic tectonic evolution and tectonic deformation resulting from the India-Asia collision (Halim et al., 1998; Cogné et al., 1999; Li et al., 2001; Chen et al., 2002; Dupont-Nivet et al., 2002, 2003; Sun et al., 2005, 2006, 2012; Fang et al., 2007; Lu et al., 2012; Zhang et al., 2013; Yu et al., 2014a, 2014b). Very little reliable Carboniferous paleomagnetic data is available from the Qaidam Block. Paleomagnetic results from only 25 samples yielded a paleolatitude of 26 °N for the Qaidam Block during the Late Carboniferous (Li et al., 1989). The Qaidam Block was considered to have been a part of the Tarim Block during the Late Carboniferous and it constituted the southern margin of Angaraland. Wu et al. (1997) considered that the Qaidam Block was located at a paleolatitude of about 12 °N and it was farther south than the Tarim Block during the late Paleozoic. However, based on the paleomagnetic results from 17 samples, Yang et al. (1992) suggested that the Qaidam Block was located at a low latitude (4.4 °N) near the equator during the Carboniferous. Recent paleomagnetic results from 56 limestone samples indicated that the Qaidam Block was located at ~13.5 °N during the Early Carboniferous (Wang et al., 2016). The foregoing review demonstrates that the earlier paleomagnetic research was hampered by an insufficient number of samples and by the lack of stability tests, such as fold test and reversal test. Hence, there are significant limitations in discussing the paleolatitude position and tectonic affinity of the Qaidam Block during the Carboniferous based on these earlier paleomagnetic studies.

In this study, we present paleomagnetic results from Early and Late Carboniferous limestones of the Qaidam Block. Our intention was to address the problem of the lack of reliable Carboniferous paleomagnetic data and thereby to provide a reliable constraint on the paleolatitude of the Qaidam Block; and to provide accurate paleomagnetic constraints on its paleogeographic reconstruction, tectonic affinity and relationship with adjacent blocks during the Carboniferous.

2. Geologic setting and sampling

The Qaidam Basin is a Cenozoic sedimentary basin underlain mainly by Precambrian crystalline basement and a Paleozoic fold belt (Wang and Coward, 1990; Zhang et al., 2008). It lies to the southeast of the Tarim Block, southwest of the Hexi Corridor-Alashan Block, and north of the Kunlun and Songpan-Ganze Blocks. It is separated from the Tarim Block by the Altyn Tagh fault zone to the north, from the Songpan-Ganze Block by the East Kunlun fault zone to the southeast, and from the Kunlun Block by the Qimen Tagh fault zone to the south (Fig. 1b). Strata outcrops in the Qaidam Basin are mainly Cenozoic sediments. Outcrops of Pre-Cenozoic strata are limited to the western and northern margins, and outcrops of Carboniferous strata are limited to the basin margin. Major Upper Carboniferous deposits in the Qaidam Basin include the Zhongwunongshan Group (C_{2zh}), the Zhabusagaxiu Formation (C_{2z}) and Keluke Formation (C_{2k}). Lower Carboniferous deposits include the Huaitoutala Formation (C_{1h}) and Chengqianggou Formation (C_{1c}). In the study area, outcropping rocks consist mainly of Paleozoic and Proterozoic strata (Fig. 1c). Examples are the Cambrian Oulongbuluke Formation, Ordovician Duoquanshan Formation, Lower Carboniferous Huaitoutala Formation, Upper Carboniferous Keluke Formation and Upper Carboniferous Zhabusagaxiu Formation. There are several NW-SE faults and left-lateral strike-slip faults in the study area (Fig. 1c).

The sampled locality lies at the northern margin of the Qaidam Block (Fig. 1). Paleomagnetic samples were collected from two different sections. A simplified geological map of the Shihuigou and Oulongbuluke sections is shown in Fig. 2. Sites MC1–MC11 from the Early Carboniferous Huaitoutala Formation and sites MC12–MC20 from the Late Carboniferous Keluke Formation were collected in the Shihuigou section (Fig. 2a). Sites TC23–TC31 from the Early Carboniferous Huaitoutala Formation were collected from the Oulongbuluke section (Fig. 2b). The Huaitoutala Formation mainly consists of biolithite limestone and marlstone intercalated with sandstone and shale. Fossils identified in this formation include anthozoa (*Yuanophy-llum knasuense Yü*, *Lithostrotion irrgulare Phillips*, *Neoclisiophyllum*, *Cravenia*, *Rylstonia*) and Brachiopoda (*Kansuella aff.kansuensis Chao*, *Gigantoproductus geniculatus*, *Echinoconchella elegans*, *Fluctuaria undata*), indicative of Early Carboniferous time (Wang, 1987; Qinghai Bureau of Geology and Mineral Resources, 1991; Chen et al., 2003). In addition, the Keluke Formation consists of biolithite limestone intercalated with quartzose sandstone. Fossils identified in the Keluke Formation include fusulinida (*Pseudostaffella* sp., *Eoparafusulina* sp.), brachiopoda (*Christites gobicus Chao*, *Echinoconchus aff.punctatus*, *Alexenia* sp.) and anthozoa (*Cystophora hunboldta*, *Caninia* sp.), indicative of Late Carboniferous time (He et al., 1987; Qinghai Bureau of Geology and Mineral Resources, 1991; Wang and Yu, 1995). The Huaitoutala Formation overlies the Chengqianggou Formation and is unconformably overlain by the Keluke Formation. The Keluke Formation is overlain by the Late Carboniferous Zhabusagaxiu Formation.

In total, we sampled 11 Early Carboniferous sites (106 samples) and 9 Late Carboniferous sites (74 samples) at the Shihuigou section (37.4 °N, 96.1 °E), and 9 Early Carboniferous sites (83 samples) at the Oulongbuluke section (37.2 °N, 96.7 °E) (Figs. 2, 3, Table 1). Sampling was mainly conducted on limestone. In general, 10 samples were taken at each site, using a gasoline-powered drill, and were oriented using a magnetic compass.

3. Laboratory techniques

In the laboratory, the samples were cut into 2.2-cm-long cylinders for subsequent paleomagnetic analysis. All samples underwent stepwise thermal demagnetization up to 580 °C, performed with an ASC TD-48 thermal demagnetizer with an internal residual field of < 10 nT. Demagnetization temperature intervals were generally large (40–60 °C) in the low temperature part, and smaller (20–30 °C) at higher temperatures. Remanent magnetizations were measured using a 2G-755R cryogenic magnetometer. In order to identify the magnetic minerals, stepwise thermal demagnetization of a three-component isothermal remanent magnetization (IRM) were measured using a JR-6 spinner magnetometer. All measurements were carried out in a shielded room with residual fields of < 300 nT at the Key Laboratory of Paleomagnetism and Tectonic Reconstruction of the Ministry of Land and Resources, CAGS, in Beijing. Magnetization directions were determined by principal component analysis (Kirschvink, 1980). The site-mean of the paleomagnetic directions was calculated using Fisher statistics (Fisher, 1953). Paleomagnetic data were analyzed using the computer program packages developed by Enkin (1990) and Cogné (2003).

4. Paleomagnetic results

Thermal demagnetization of a three-component isothermal remanent magnetization (IRM) (Lowrie, 1990) was conducted on representative samples of the Huaitoutala Formation in the Shihuigou and Oulongbuluke sections. Fields of 1.2 T, 0.4 T and 0.12 T were used to characterize the hard, medium and soft components, respectively. The results indicate that the hard components (0.4–1.2 T) are unblocked at about 680 °C (Fig. 4a–b). Thermal demagnetization of the soft (< 0.12 T) and medium (0.12–0.4 T) components indicates an

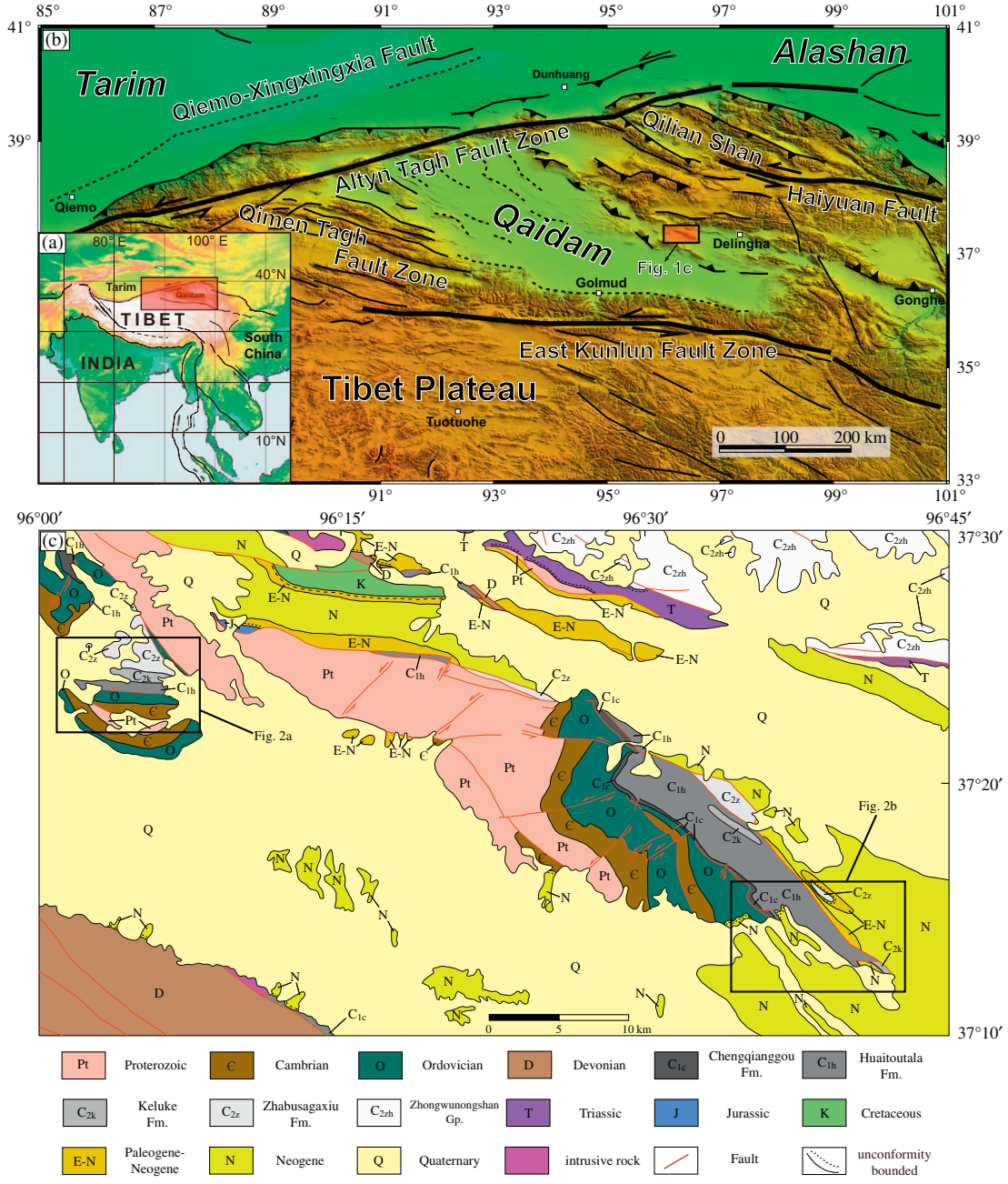


Fig. 1. Tectonic sketch map of Central Asia (a), regional geological map of the Qaidam Block and adjacent areas (b), simplified geological map of the study area (c).

unblocking temperature of 580 °C (Fig. 4a–b). This finding indicates that magnetite and hematite are the main magnetic carriers in the limestone samples in the studied section. However, hematite is not visible in the demagnetization diagrams (Fig. 5). The reason for this phenomenon is likely because that the magnetic intensity of hematite in the limestone samples are too weak to contribute for the remanent magnetization.

4.1. Lower Carboniferous Huaitoutala Formation

Fig. 5a–f shows representative Zijderveld plots for the Huaitoutala Formation. In most samples, two components could be separated: a lower-temperature component (LTC) which was erased before 240 °C, and a higher-temperature component (HTC) which was isolated between 300 °C and 510 °C. The LTC is close to the present geomagnetic field (PGF) direction (PGF: D/I = 359.6°/57.2°; LTC: Dg/Ig = 2.1°/51.0° with α_{95} = 5.9°) (Fig. S1), and thus it should be a viscous remanent magnetization of the PGF direction. The HTC directions in geographic coordinates are significantly different from the present-day field direction (Fig. 6). Sites TC23, TC29, TC30, TC31 have $\alpha_{95} > 19°$ (indicated by asterisks in Table 1) and therefore we excluded them from the calculation of the site-mean direction. The site mean HTC direction of 15 sites is Dg = 311.7°, Ig = −22.9°, kg = 6.0, α_{95} = 17.0° (in geographic coordinates), and Ds = 308.7°, Is = −48.2°, ks = 22.3, α_{95} = 8.3° (after tectonic correction); the corresponding paleopole lies at −8.0°N, 139.9°E with A_{95} = 8.8°.

In Table 1 and Fig. 6a, the declination of the site-mean directions from sites TC25–28 indicates a difference of 40° relative to those from sites MC1–11. This may be due to oroclinal bending between the studied areas (Fig. 1c). The Paleozoic formations in the Shihuigou section (Fi. 2a) and studied area (Delingha) trend in an E-W direction (Geological map of Qinghai Province of the People's Republic of China).

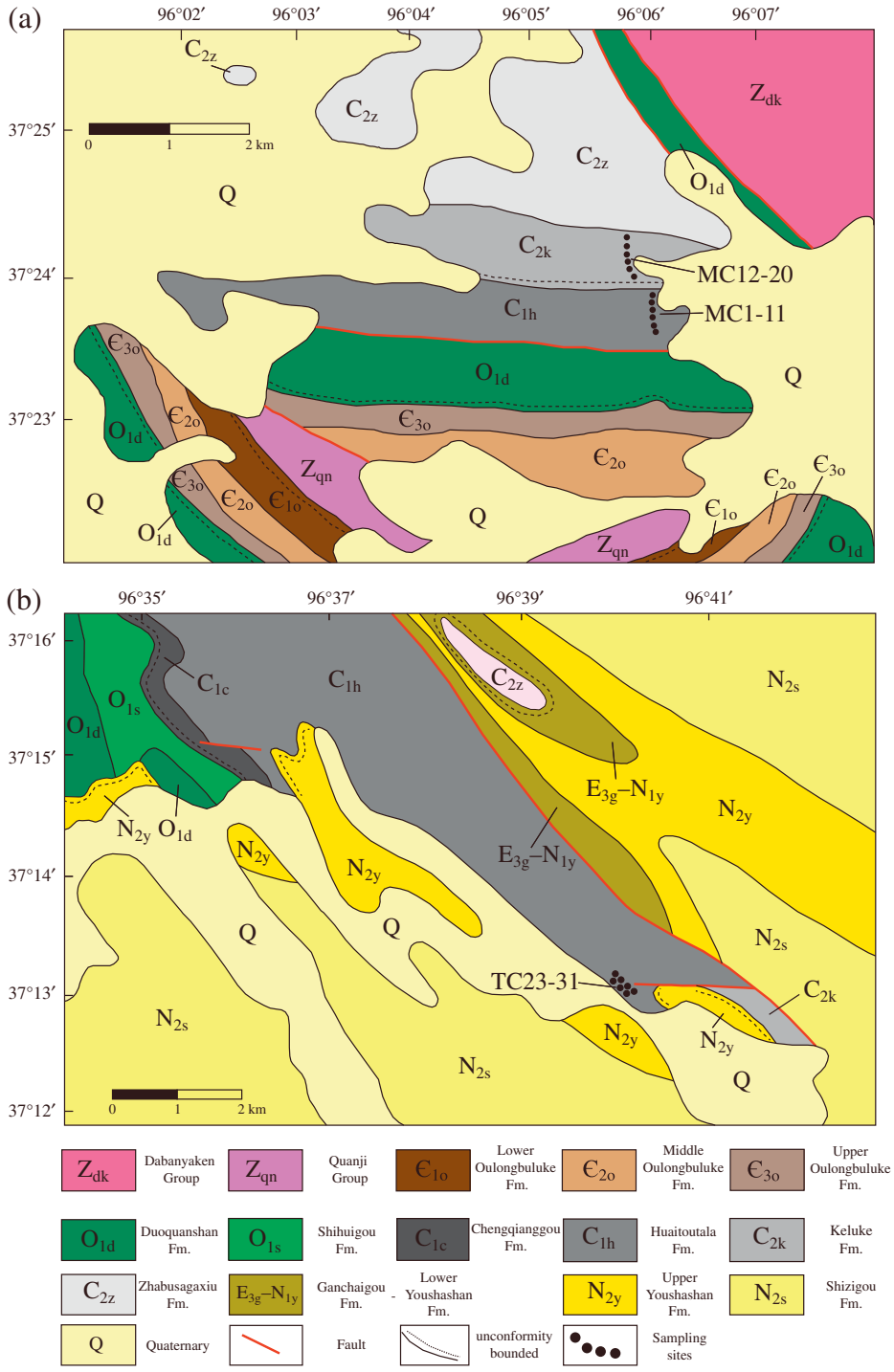


Fig. 2. Simplified geological map of the sampling area. (a) Shihuigou section, (b) Oulongbuluke section. Solid circles show paleomagnetic sampling locations.

However, the Paleozoic structures in the Oulongbuluke section (Fi. 2b) trend in a NW-SE direction. This indicates that the Paleozoic strata in the Oulongbuluke area may be influenced by a local orocline. After anticlockwise rotation of the site-mean directions for sites TC25–28 from the Oulongbuluke section, the site mean HTC direction of 15 sites is $Ds = 298.6^{\circ}$, $Is = -46.4^{\circ}$, $ks = 66.4$, $\alpha_{95} = 4.7^{\circ}$ (after tectonic correction); the corresponding paleopole lies at $-3.2\,^{\circ}$N, $147.5\,^{\circ}$E with $A_{95} = 4.8^{\circ}$. It passes both the McElhinny (1964) and McFadden (1990) fold tests at the 95% and 99% confidence levels (in situ Xi1 = 13.620, Xi2 = 13.680; tilt-corrected Xi1 = 0.9784, Xi2 = 2.980; Xi = 5.583 critical value at 95% confidence, Xi = 7.836 critical value at 99% confidence). A progressive unfolding correction shows that the site-mean direction has the highest precision parameter at 92.8% unfolding (with 95% uncertainties ranging from 88% to 97.9%) (Fig. 6c) (Watson and Enkin, 1993). In addition, the tilt-corrected Fisher precision parameter (k) increases significantly from 16.4 to 66.4 (Fig. 6b, Table 1). The inclination-only mean of 15 site-mean directions is $-46.0^{\circ} \pm 5.0^{\circ}$. This is highly consistent with the inclination of $-46.4^{\circ} \pm 4.7^{\circ}$ of the Fisher site-mean direction. Therefore, the site-mean result obtained from the Huaitoutala Formation can be used to constrain the paleolatitude of the Qaidam Block during the Early Carboniferous.

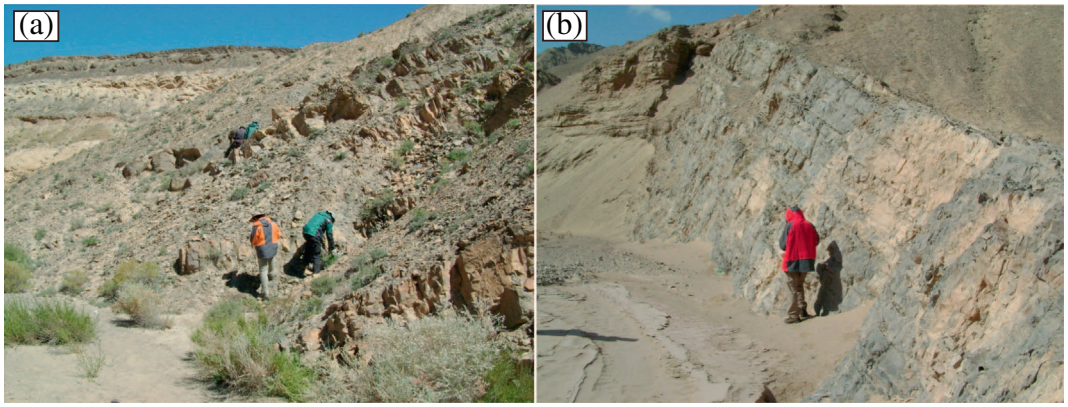


Fig. 3. Photographs showing field outcrops and sampling sites of the Carboniferous limestones in (a) Shihuigou section and (b) Oulongbuluke section.

4.2. Upper Carboniferous Keluke Formation

Representative Zijderveld plots for the Keluke Formation are shown in Fig. 5g–i. Most of the specimens have a LTC between the NRM and 240 °C, which is probably a viscous remanent magnetization of the PGF direction (Fig. S1). The HTC direction could be isolated between 300 °C and 510 °C. The site-mean directions of the Keluke Formation are listed in Table 1 and illustrated in Fig. 7. The HTC directions in geographic coordinates are significantly different from the present-day field direction. Because site MC18 has $\alpha_{95} > 20^{\circ}$ we excluded it from

Table 1
Site-mean paleomagnetic directions of the High-temperature from the Early and Late Carboniferous sediments in the Qaidam Block.

Site	Slat. (°N)	Slon. (°E)	Strike/dip (°)	n/N	D_g (°)	I_g (°)	D_s (°)	I_s (°)	κ	α_{95} (°)
Lower Carboniferous Huaitoutala Fm.										
MC1	37.3942	96.0987	272/64	8/9	325.8	− 9.6	290.9	− 52.0	32.7	9.8
MC2	37.3942	96.0987	272/64	9/10	324.6	− 9.8	289.5	− 51.1	46.3	7.6
MC3	37.3948	96.0987	272/53	5/5	321.3	− 8.8	299.1	− 43.6	40.9	12.1
MC4	37.3948	96.0987	277/61	10/10	318.2	− 3.7	296.2	− 37.3	28.2	9.3
MC5	37.3952	96.0985	277/61	7/8	323.9	− 14.1	288.2	− 47.5	127.2	5.4
MC6	37.3952	96.0985	277/61	8/8	321.7	− 13.5	287.4	− 45.4	27.3	10.8
MC7	37.3952	96.0985	278/49	8/8	330.9	− 13.2	307.8	− 47.4	27.4	8.9
MC8	37.3952	96.0985	280/55	10/10	327.9	− 18.9	292.2	− 49.5	16.8	12.2
MC9	37.3952	96.0985	276/46	8/8	331.1	− 18.9	305.5	− 51.5	41.5	8.7
MC10	37.3952	96.0985	276/46	7/7	324.4	− 14.2	302.9	− 43.8	27.9	11.6
MC11	37.3957	96.0986	276/46	7/8	327.2	− 12.4	307.4	− 44.2	21.4	13.3
TC23[a]	37.2187	96.6652	126/78	4/7	250.5	− 31.4	330.4	− 46.2	23.5	19.4
TC25	37.2187	96.6652	115/78	6/7	231.0	− 21.3	318.9	− 63.5	37.1	11.1
TC26	37.2187	96.6652	115/78	7/7	248.7	− 40.8	340.2	− 42.2	20.5	13.7
TC27	37.2187	96.6652	115/78	3/4	249.0	− 44.6	344.6	− 40.3	726.3	4.6
TC28	37.2187	96.6652	121/65	5/7	281.9	− 58.3	355.6	− 31.0	47.0	11.3
TC29[a]	37.2200	96.6666	3/35	4/7	21.0	− 42.6	346.9	− 43.3	23.2	19.5
TC30[a]	37.2200	96.6666	3/35	4/7	5.4	− 32.2	344.9	− 27.2	21.1	20.5
TC31[a]	37.2200	96.6666	3/35	5/5	13.9	− 68.7	342.6	− 34.1	7.3	30.3
Mean	37.3	96.4		15(108)	311.7	− 22.9	–	–	6.0	17.0
					–	–	308.7	− 48.2	22.3	8.3
Corresponding pole: − 8.0 °N/139.9 °E, A_{95} = 8.8°										
Mean[b]	37.3	96.4		15(108)	308.7	− 15.0	–	–	6.4	16.4
					–	–	298.6	− 46.4	66.4	4.7
Corresponding pole: − 3.2 °N/147.5 °E, A_{95} = 4.8°										
Upper Carboniferous Keluke Fm.										
MC12	37.4004	96.0969	276/46	9/9	339.4	− 1.7	332.1	− 45.3	13.3	14.6
MC13	37.4004	96.0969	276/46	10/10	332.6	− 4.0	323.5	− 40.9	116.7	4.5
MC14	37.4004	96.0969	276/46	8/8	321.4	5.3	319.2	− 43.4	45.3	8.3
MC15	37.4004	96.0969	276/46	8/10	330.0	1.9	331.4	− 47.0	54.3	7.6
MC16	37.4004	96.0969	266/46	5/5	335.4	− 7.5	322.0	− 50.6	61.2	9.9
MC17	37.3845	96.0963	266/46	6/7	329.2	− 4.9	315.5	− 45.4	27.9	12.9
MC18[a]	37.3845	96.0963	275/57	3/5	329.1	1.4	313.2	− 41.8	38.0	20.3
MC19	37.3845	96.0963	275/57	5/6	328.2	2.0	312.9	− 40.7	42.8	11.8
MC20	37.3845	96.0963	279/63	10/12	334.9	− 8.0	303.0	− 52.5	26.0	9.7
Mean	37.4	96.1		8(61)	331.4	− 2.1			123.4	5.0
							320.1	− 46.1	108.2	5.3
Corresponding pole: − 15.1 °N/132.2 °E, A_{95} = 5.4°										

Notes: Slat. and Slon.: latitude and longitude of the sampling site; n/N: number of samples used to calculate mean and measured; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; κ: the best estimate of the precision parameter; α_{95}: the radius that the mean direction lies within 95% confidence. Strike: dip direction of beds − 90°, Dip: the mean dip azimuth of site beds.

[a] Sites were not used to calculate the final mean direction.

[b] Site-mean directions after a 40° anticlockwise rotation of the sites TC25–28.

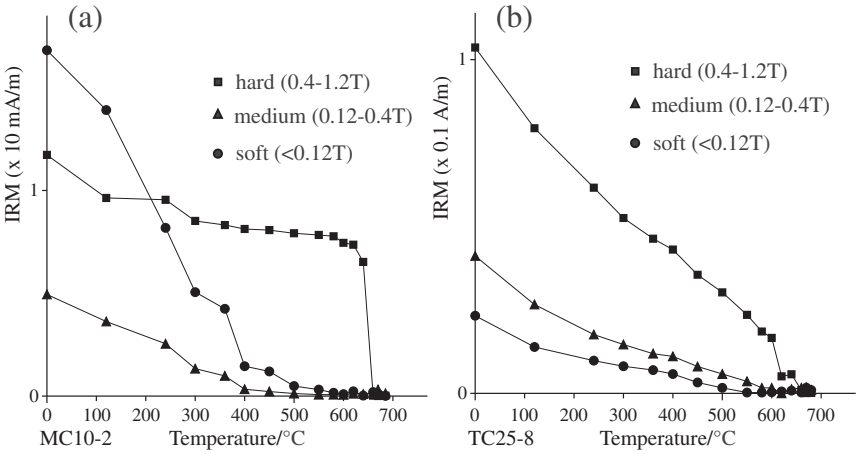


Fig. 4. Results of thermal demagnetization of a three-component IRM for representative samples. The variation of soft (< 0.12 T), intermediate (0.12–0.4 T), and hard (0.4–1.2 T) components are shown during thermal demagnetization.

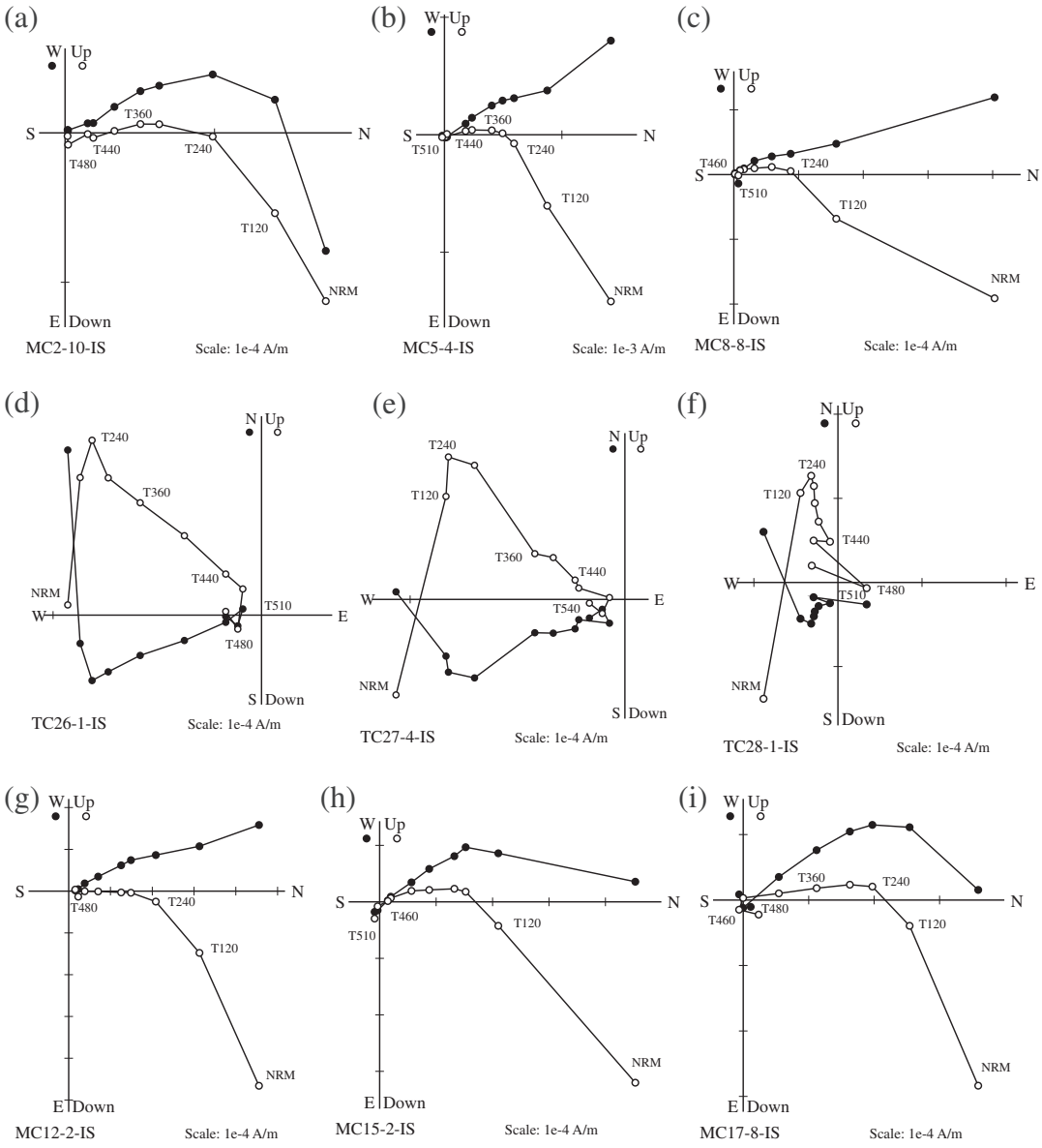


Fig. 5. Orthogonal demagnetization diagrams of representative samples (sample number and corresponding sites are listed) in situ coordinates. Demagnetization steps are in °C in all plots. NRM is the natural remanent magnetization. (a–f) Typical examples from Early Carboniferous limestone; (g–i) typical examples from Late Carboniferous limestone.

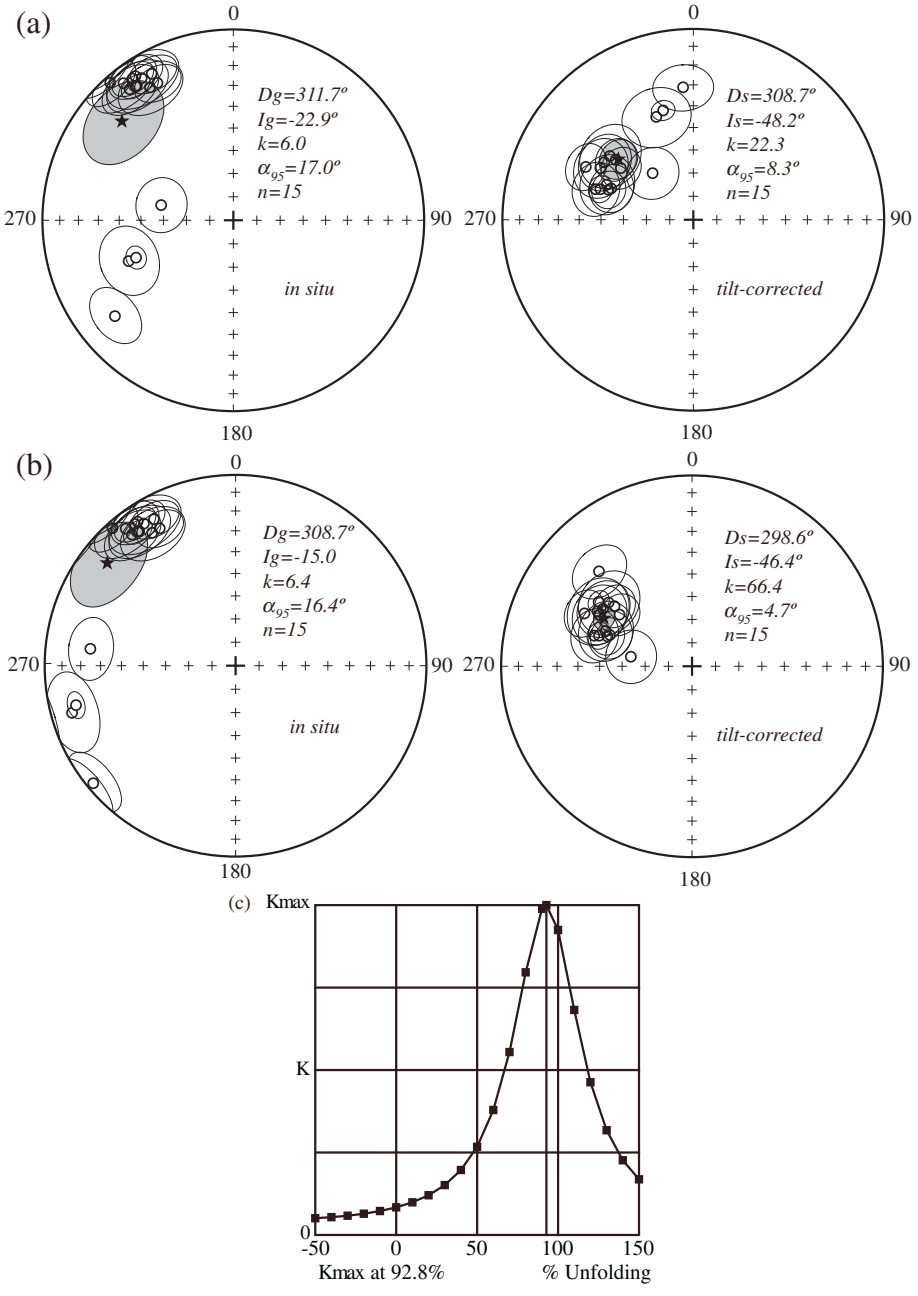


Fig. 6. Equal-area projections of site-mean directions from the Early Carboniferous limestone (a) and after a 40° clockwise rotation of the sites TC25–28 (b), and the progressive unfolding analysis of the HTC directions using the Watson and Enkin (1993) method (c). The stars in (a) and (b) indicate the overall-mean direction of 15 sites.

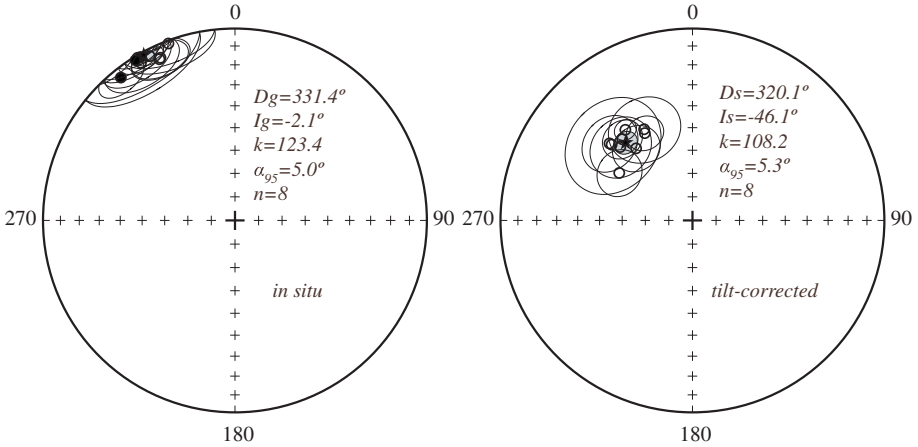


Fig. 7. Equal-area projections of site-mean directions from the Late Carboniferous Keluke Fm. limestones. The stars indicate the overall-mean direction of 8 sites.

calculation of the site-mean direction. The site mean HTC direction of 8 sites is $Dg = 331.4^{\circ}$, $Ig = -2.1^{\circ}$, $kg = 123.4$, $\alpha_{95} = 5.0^{\circ}$ (in geographic coordinates), and $Ds = 320.1^{\circ}$, $Is = -46.1^{\circ}$, $ks = 108.2$, $\alpha_{95} = 5.3^{\circ}$ (after tectonic correction); the corresponding paleopole lies at $-15.1\ ^{\circ}$N, 132.2 °E with $A_{95} = 5.4^{\circ}$.

Because of the monoclinal bedding attitude in the studied section, no fold test can be applied to the paleomagnetic directions from the Upper Carboniferous limestone. Nevertheless, the HTC directions are different from the younger results and PGF for the Qaidam Block (Late Permian: 333.1°/43.0°, $\alpha_{95} = 6.0^{\circ}$, Xu et al., 2011; Late Jurassic: 42.6°/38.4°, $\alpha_{95} = 8.0^{\circ}$, Chen et al., 2002; Early Cretaceous: 17.4°/55.3°, $\alpha_{95} = 4.5^{\circ}$, Sun et al., 2006; Eocene: 8.0°/42.4°, $\alpha_{95} = 5.2^{\circ}$, Dupont-Nivet et al., 2002; Neogene: 0.5°/48.7°, $\alpha_{95} = 3.6^{\circ}$, Dupont-Nivet et al., 2002; PGF: 359.7°/56.8°). Moreover, the HTC directions are all reversed polarity and are consistent with the Kiaman reversed superchron (KRS). Hence, the paleomagnetic direction from the Early and Late Carboniferous rocks can be regarded as the primary magnetization.

5. Discussion

5.1. Paleolatitude of the Qaidam Block during the Carboniferous

There are only a few studies of the paleo-latitudinal location of the Qaidam Block during the Carboniferous. According to the dynamic evolution of the Paleozoic geography of eastern Asian, Cocks and Torsvik (2013) considered that the Qaidam Block was located at about 0 °S and 10 °S during the Early and Late Carboniferous, respectively. However, the most recent global paleogeographic reconstruction shows that the Qaidam Block was located at about 30 °N during the Carboniferous (Domeier and Torsvik, 2014). The differences in the paleogeographic reconstruction of the Qaidam Block during the Carboniferous may be the result of the sparsity of paleomagnetic data. Previous paleomagnetic studies indicated that the Qaidam Block was located at a low latitude during the Carboniferous (Yang et al., 1992; Wu et al., 1997; Wang et al., 2016). Firstly, the paleomagnetic data reported by Yang et al. (1992) and Wu et al. (1997) lack a stability test. In addition, the previous Carboniferous paleomagnetic data are all based on insufficient samples, and it was not possible to apply the elongation/inclination (E/I) correction method to evaluate inclination shallowing. Hence, the reliability of the previous paleomagnetic data needs further validation.

In this study, we have obtained the primary magnetization from Early and Late Carboniferous limestones in the Qaidam Block. In addition, to examine the influence of inclination shallowing, we applied the E/I correction method (Tauxe and Kent, 2004) for the Early Carboniferous paleomagnetic result. The E/I correction shows that the corrected inclination for the Early Carboniferous ranges from 44.5° to 58.3°, and the original inclination (−46.4°) falls within the range of this confidence interval (Fig. S2). This suggests that the Carboniferous sediments in the studied area have not been affected by significant inclination shallowing. Consequently, the new paleomagnetic results are more suitable for paleogeographic reconstructions of the Qaidam Block during the Carboniferous.

The paleolatitude of the Qaidam Block during the Early and Late Carboniferous is 27.7° ± 4.7 °N and 27.7° ± 5.2 °N (reference point: 37.3 °N, 96.4 °E), respectively. This finding implies that the Qaidam Block experienced insignificant northward movement between the Early and Late Carboniferous. However, the paleomagnetic results indicate a significant difference of 21.8° in declination suggesting that the Qaidam Block experienced a counter-clockwise rotation of 21° between the Early and the Late Carboniferous.

5.2. Tentative paleogeographic reconstruction

In order to better understand the kinematic relationships between the Qaidam Block and adjacent blocks, and to reconstruct the paleogeography of the major blocks of Central Asia, previous Early and/or Late Carboniferous paleomagnetic results for the Qaidam Block and its surrounding blocks are summarized as follows.

The Qaidam and Qilian Blocks can be combined as a composite terrane during the post-early Paleozoic based on the evidence of ultrahigh pressure metamorphism (UHPM) from north the Qaidam Block (Shi et al., 2006; Wu et al., 2007; Zhang et al., 2006, 2008; Chen et al., 2009; Xiao et al., 2009; Xiong et al., 2011; Song et al., 2012, 2014). The Carboniferous paleolatitude of the Qaidam Block was determined by this study. The Late Carboniferous paleomagnetic pole of the Tarim Block has been documented previously (Bai et al., 1987; Fang et al., 1996; Gilder et al., 1996). However, the paleomagnetic data calculated from Gilder et al. (1996) was a Permian remagnetization and the results reported by Bai et al. (1987) lacked a robust field test to assess data reliability. The paleomagnetic pole (46.6 °N, 170.2 °E with $A_{95} = 4.4^{\circ}$) reported by Fang et al. (1996), with sufficient samples and a robust field test, is accepted for the Tarim Block. Recent paleomagnetic evidence from Inner Mongolia indicates that the Paleo-Asian Ocean has been closed since at least the Late Devonian (Zhao et al., 2013), and thus the Late Carboniferous Inner Mongolian pole (Zhao et al., 1990) can be used for the North China Block (Yuan and Yang, 2015). Both geochronologic and paleomagnetic studies indicate that the Alashan Block was separated from the North China Block prior to the Mesozoic (Yang et al., 2014; Yuan and Yang, 2015). Hence, the Hexi Corridor-Alashan Block can be treated as an independent block during the late Paleozoic. The Early and Late Carboniferous paleomagnetic poles for the Hexi Corridor-Alashan Block were reported by Huang et al. (2001) and Yuan and Yang (2015), respectively. The Early and/or Late Carboniferous paleomagnetic data for the Qaidam, Tarim, North China, South China, Junggar, Yili, and Kazakhstan-Tianshan Blocks are listed in Table 2. The paleogeographic position of Siberia was reconstructed from its Paleozoic apparent polar wandering path (APWP) (Metelkin et al., 2012).

The paleogeographic reconstruction of the Qaidam Block and the other major blocks of Central Asia during the Early and Late Carboniferous are shown in Fig. 8. The paleogeographic reconstruction results show that South China remained in the proximity of the paleoequator during the Carboniferous (Fig. 8). The Siberia, Kazakhstan-Tianshan and Hexi Corridor-Alashan Blocks were in a stable position between the Early and Late Carboniferous. The Qaidam, Tarim and Kazakhstan-Tianshan Blocks were all located at around 30 °N during the Carboniferous. However, the significant difference in paleolatitude (at the same reference point) between the Qaidam and the Tarim Blocks during the Late Carboniferous indicates that they may not have accreted together by that time.

5.3. Tectonic implications

Fig. 9 shows a graph of the paleo-latitudinal variation of the major blocks of Central Asia, from the Carboniferous to the Triassic. The similar paleolatitude for the Tarim, Junggar and Kazakhstan-Tianshan Blocks indicates that they may have been in close proximity since the Late Permian. Recent paleomagnetic results imply that the Qaidam Block was located at about 25 °N during the Late Permian (Xu et al., 2011). The paleo-position of the Tarim Block is slightly further north than that of the Qaidam Block during the Late Permian (Fig. 9), and hence the amalgamation of the Qaidam and Tarim Blocks may have taken place after the Late Permian. Comparison of the Late Permian and Carboniferous paleolatitude indicates that the Qaidam Block experienced insignificant latitudinal movement during the late Paleozoic. The Tarim, North China, Hexi Corridor-Alashan and Qaidam Blocks are located to the south of the Central Asian Orogenic Belt, and they are all included in the Tarim–North China collage system (Xiao et al., 2015). The formation of the Tarim–North China collage system was mainly completed during the end-Permian to the Middle Triassic (Xiao et al., 2015). In contrast, the paleolatitude of the Qaidam Block was further

Table 2
Available Carboniferous to Triassic paleomagnetic poles for the Qaidam Block and its adjacent blocks.

Blocks	Age	Slat. (°N)	Slon. (°E)	N(n)	Plat. (°N)	Plon. (°E)	A_{95} (dp/dm)	Paleolatitude (°N)	Test	P ± ΔP (°)	Reference
Qaidam	P_2	37.5	99.1	9(56)	64.5	345.4	5.9	25.1 ± 5.9	R	17.3 ± 7.9	Xu et al. (2011)
	C_2	37.4	96.1	8(61)	−15.1	132.2	5.4	27.7 ± 5.4	No	11.9 ± 7.1	This study
	C_1	37.3	96.4	15(126)	−3.2	147.5	4.8	27.7 ± 4.8	F	13.2 ± 6.2	
Tarim	T_3			5[a]	57.3	169.9	8.1	39.2 ± 8.1	–	24.9 ± 6.7	Zhao et al., 2014
	P_2			5[a]	69.9	188.5	5.4	34.0 ± 5.4	–	8.4 ± 7.7	
	P_1			5[a]	59.7	175.1	7.1	37.0 ± 7.1	–	6.5 ± 8.4	
	C_2	39.9	80.2	8(−)	46.6	170.2	4.4	36.4 ± 3.6	F	3.3 ± 6.7	Fang et al. (1996)
North China	C_2	43.9	120.1	3(29)	32.6	338.1	4.4	0.5 ± 6.2	No	39.2 ± 6.7	Zhao et al. (1990)
South China	C_2	25.8	107.5	−(7)	53.5	229.9	12.1	9.3 ± 12.1	No	30.4 ± 10.6	Zhang et al. (2001)
	C_1	25.9	107.4	−(24)	47.5	229.1	9.6	4.7 ± 9.6	No	36.2 ± 8.7	
Hexi Corridor-Alashan	T_{1-2}	37.2	104.4	24(196)	39.6	12.7	5.2	27.0 ± 5.2	F,R	20.2 ± 7.0	Liu et al. (2010)
	P_2	39.2	106.3	9(68)	27.2	18.8	12.0	25.4 ± 12.0	F	16.9 ± 11.0	Yuan and Yang (2015)
	C_2	37.0	105	7(48)	13.1	11.0	7.0	11.5 ± 7.0	F	28.1 ± 7.8	
	C_1	37.2	105.4	10(39)	10.5	14.0	6.2	12.3 ± 6.2	F	28.6 ± 6.9	Huang et al. (2001)
Junggar	T_3	43.8	89.2	6(51)	87.5	106.2	13.1	39.8 ± 13.1	No	24.3 ± 10.1	Choulet et al. (2013)
	T_1	43.0	87.5	7(39)	68.0	3.0	14.6	33.0 ± 14.6	R	14.1 ± 12.2	
	P_2	43.8	87.8	6(43)	60.4	4.7	5.4	31.0 ± 5.4	F	11.4 ± 7.7	Choulet et al. (2011)
	P_1	43.8	87.8	4(30)	79.5	36.6	6.8	42.0 ± 6.8	F	1.4 ± 8.3	
	C_2	46.5	83	12(107)	70.7	209.0	5.0	28.1 ± 5.0	F	11.6 ± 6.9	Yi et al. (2015)
Yili	P_2	43.4	82.5	5(30)	79.7	172.0	11.3	39.2 ± 11.3	No	3.1 ± 10.6	Wang et al. (2007)
	C_2	43.3	83.3	15(98)	68.6	290.6	6.1	16.5 ± 6.1	F	23.2 ± 7.4	
North Tien Shen	P_2	41.8	73.5	–	73.6	200.8	4.2	31.7 ± 4.2	–	9.2 ± 6.0	Bazhenov et al. (2003)
	C_2	42.9	71.7	9(49)	53.8	202.1	4.2	21.2 ± 4.2	F	19.7 ± 6.0	
	C_1	41.7	74.3	16(138)	−63.3	150.8	5.8/3.9	19.5 ± 4.8	F	21.4 ± 6.2	Kirscher et al. (2014)
Siberia (APWP)	T_3				47.7	128.8	4.3	64.2 ± 4.3			Metelkin et al. (2012)
	T_2				52.0	155.0	8.0	47.1 ± 8.0			
	P_2				46.0	161.0	9.0	42.3 ± 9.0			
	P_1				42.0	158.0	9.0	43.4 ± 9.0			
	C_2				35.0	160.0	8.0	39.6 ± 8.0			
	C_1				22.0	151.0	7.0	40.9 ± 7.0			

Notes: Slat. and Slon., latitude and longitude of the sampling site; N(n): number of sites (samples); Plat. and Plon., latitude and longitude of the paleomagnetic pole; A_{95}, the radius that the mean pole lies within 95% confidence; Paleolatitude, paleolatitude calculated for the reference point at 37.3 °N, 96.4 °E; F, positive of fold test; P ± ΔP, the paleo-latitudinal difference with 95% confidence limit of the corresponding block with respect to Siberian craton.

[a] Denotes the number of paleomagnetic poles used.

south than the Tarim Block prior to the Late Permian, and hence it may have experienced a relatively rapid northward movement and collided with the Tarim Block after the Late Permian (Fig. 9).

Compared with the Late Permian paleomagnetic pole of the Alashan Block (Yuan and Yang, 2015) and the Qaidam Block (Xu et al., 2011), the two blocks were in close proximity during the Late Permian. However, they were far apart during the Carboniferous (Fig. 9). The paleomagnetic data from the Hexi Corridor Block suggests that it was not connected with the Tarim Block by the Early-Middle Triassic (Liu et al., 2010). Hence, the Hexi Corridor-Alashan Block may have been connected with the Qaidam Block during the Permian, and they amalgamated with the Tarim Block after Middle Triassic.

6. Conclusions

We have reported new paleomagnetic results from Early and Late Carboniferous sediments from the Qaidam Block. Systematic thermal demagnetization yielded a HTC magnetic component for the Early Carboniferous and Late Carboniferous sediments, which passed the fold test and is consistent with the Kiaman reversed superchron (KRS), respectively. These findings suggest that the HTC directions can be regarded as the primary magnetization. The new paleomagnetic results indicate that the Qaidam Block was located at about 27 °N and experienced no paleomagnetically-detectable latitudinal movement during the Carboniferous. However, it experienced a counter-clockwise

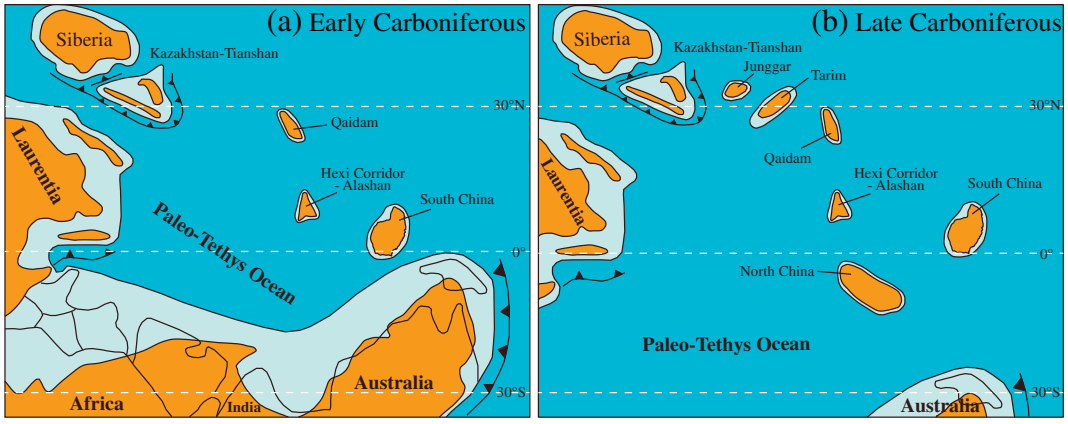


Fig. 8. Schematic maps showing paleogeographic reconstruction of the Qaidam Block and other major blocks of Central Asia during (a) the Early and (b) the Late Carboniferous (modified from Metcalfe, 2013). Relative longitudes of the blocks are unconstrained. The reference paleomagnetic poles are listed in Table 2.

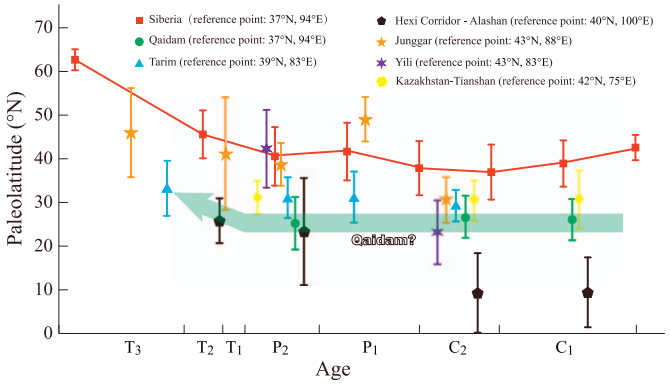


Fig. 9. Paleolatitude plots for the major blocks of Central Asia. The vertical bars show the errors of the paleolatitudes. The paleolatitudes for the small blocks are calculated respectively by a reference site within the respective block. Red squares: the Siberian continent from Metelkin et al. (2012), green circles: the Qaidam Block (C1, C2-this study; P2-Xu et al., 2011), blue triangles: the Tarim Block (C2-Fang et al., 1996; P1, P2, T3-Zhao et al., 2014), black pentagons: the Hexi Corridor-Alashan Block (C1-Huang et al., 2001; C2, P2-Yuan and Yang, 2015; T1–2-Liu et al., 2010), orange stars: the Junggar Block (C2-Yi et al., 2015; P1, P2-Choulet et al., 2011; T1, T3-Choulet et al., 2013), PURPLE hexagrams: the Yili block (Wang et al., 2007), yellow hexagons: the North Tien Shan Block (C1-Kirscher et al., 2014; C2, P2-Bazhenov et al., 2003). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

rotation of 21° during this period. Comparisons of the paleomagnetic results of the Qaidam and Tarim Blocks suggest that the Qaidam Block may have experienced a relatively rapid northward movement and collided with the Tarim Block after the Late Permian. Based on the paleomagnetic poles of the Qaidam Block and surrounding blocks, we also present a tentative paleogeographic reconstruction of Central Asia during the Carboniferous.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at http://dx.doi.org/10.1016/j.tecto.2017.08.014.

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